

Deep Learning for Smart Surveillance: Multi-Class Detection of People, Weapons, and Masks using YOLOv11

Ludovica Beritelli¹, Roberta Avanzato², Luca Guarnera¹, Francesco Beritelli², Sebastiano Battiato¹

¹ Department of Mathematics and Computer Science, University of Catania, Catania, Italy

Emails: ludovica.beritelli@phd.unict.it, luca.guarnera@unict.it, sebastiano.battiato@unict.it

² Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy

Emails: roberta.avanzato@unict.it, francesco.beritelli@unict.it

Abstract — Computer vision technologies are revolutionizing the field of security, providing intelligent and automated tools for crime prevention. In this study, we propose an advanced approach for the simultaneous detection of people, weapons, and facial masks in images extracted from surveillance camera video streams. Unlike most works in the literature, which focus on static and less contextualized scenarios, our method operates on real-world, dynamic scenes using the YOLOv11 architecture. Experimental results show an accuracy of 65.35% on the test set, with a weighted precision of 71.94%, demonstrating the model's ability to reduce false positives. This work represents a concrete step towards intelligent monitoring systems, with promising improvements achievable through dataset optimization and the integration of advanced deep learning techniques, aiming to enhance public security effectiveness.

Keywords — *Computer Vision, Deep Learning, Object Detection, YOLO, Surveillance Video, Weapon Detection, Person Detection, Face Mask Recognition*

I. INTRODUCTION

In recent years, systems for citizen monitoring and security across various application domains have gained increasing importance. A key role in the development of such solutions is played by Computer Vision (CV), which is capable of analyzing data from public and/or private video sources, such as surveillance systems. Nowadays, numerous cameras are distributed across various locations to ensure video surveillance; however, they are often utilized or analyzed only after an incident has occurred. Furthermore, human supervision is generally required to identify individuals or determine the cause of incidents, making surveillance cameras essentially passive tools. In this context, cities have been witnessing a growing number of criminal events during public gatherings, in parks, and within private venues. Several scientific studies have proposed methods to automatically detect various incidents, such as theft [1]–[3], arson, fights, abuse, disruptive behavior [4]–[8], riots between police and civilians [9], and the possession of firearms or bladed weapons [4]. While crime cannot be predicted before it occurs, the likelihood of criminal acts can be assessed by carefully

observing suspicious behavior and the surrounding environment, leveraging surveillance cameras supported by Machine Learning (ML) and Artificial Intelligence (AI) techniques [10].

Several studies have explored image segmentation and object detection. Furthermore, the importance of adaptive and high-performance systems has also been highlighted in the context of cybersecurity threat detection [11], reinforcing the relevance of robust architectures for real-time analysis under varying operational conditions.

Among the existing architectures, YOLO stands out as one of the most advanced object detection frameworks available today, known for its speed and accuracy, and enabling effective object detection even in crowded or dynamic scenes [12]. This makes it particularly suitable for applications requiring real-time analysis. The structure of this paper is as follows: Section II reviews the state of the art in the automatic recognition of weapons and people. Section III details the proposed method, including dataset construction and the configuration of the YOLOv11 architecture used for detection. Section IV presents the experimental results with a quantitative evaluation of the model's performance. Section V provides an in-depth discussion of the obtained results, highlighting the main limitations of our approach. Finally, Section VI discusses potential future developments and planned improvements.

II. WEAPON AND PERSON RECOGNITION IN SECURITY CONTEXTS

Traditional methods for weapon detection, based on manual inspections and continuous visual monitoring of surveillance footage, are often ineffective due to the high human workload, significant false positive rates, and natural susceptibility to errors. In response to these limitations, there is a growing demand for automated solutions leveraging computer vision techniques, capable of identifying suspicious activities, such as armed robberies, in real-time from CCTV video streams. Several recent studies have highlighted the effectiveness of deep learning in overcoming the shortcomings of traditional approaches.

In [13], the authors emphasize key limitations of current systems, including the identification of specific weapon types, handling of partially visible objects, and challenges in detecting small-sized items. Deep learning models have demonstrated superior accuracy and speed; however, their real-time application remains challenging due to high computational demands and the need for large, balanced datasets.

In [14], a comparison between YOLOv4 and R-CNN for weapon detection in surveillance videos shows that YOLOv4 is more efficient, achieving a mAP of 96.04% and an average accuracy of 73%, making it suitable for real-time applications. Similarly, [15] demonstrates that YOLOv8 achieves high performance in distinguishing weapons from non-weapons, even in complex environments such as schools and airports.

In [16], a Scaled-YOLOv4 model trained on a custom dataset achieves a mAP of 92.1%, offering a good trade-off between accuracy and inference time across both high-performance GPUs (e.g., RTX 2080Ti) and edge devices (e.g., Jetson Nano), confirming its applicability in real-world scenarios. Other approaches combine weapon detection with behavioral analysis.

Mukto et al. [17] propose a multi-model system integrating YOLOv5 for weapon detection and MobileNetV2 for face and violent behavior recognition, demonstrating effectiveness in realistic scenarios and timely alarm generation.

An additional advancement is presented in [18], where the system not only detects the presence of weapons but also analyzes the individuals' posture to infer weapon usage context. Similarly, Ruiz-Santaquiteria et al. [19] propose an architecture combining pose keypoints and hand bounding boxes, making robust decisions through independent CNNs. Their work has been further extended in [20], [21] using advanced data augmentation techniques and Distilled Data-efficient Image Transformers (DeiT), showing strong potential for multimodal context analysis.

Unlike most studies in the literature, which focus on isolated or poorly contextualized images, our approach uses a dataset constructed from real-world surveillance video streams. This enables the capture of complex and realistic scenes, featuring multi-class object detection (people, weapons, masks) under varying lighting and quality conditions. Table I provides an overview of recent studies focused on weapon and person detection in surveillance scenarios, highlighting the models used, data sources, and the types of objects or actions detected. Currently, no specific studies have applied YOLOv11 to this task. However, the evidence from previous versions (YOLOv5–YOLOv8) suggests that the new version could offer improved accuracy, speed, and robustness, particularly in dynamic and complex contexts like those analyzed here. Therefore, the proposed approach represents an original and potentially significant contribution to the field

Table I. DETAILS WITH EXISTING STUDIES ON WEAPON/PERSON DETECTION IN SURVEILLANCE CONTEXTS

Ref	Model	Data Type	Detected classes
[13]	Deep learning	Custom dataset	Weapons
[14]	YOLOv4, R-CNN	Surveillance videos	Weapons
[15]	YOLOv8	Simulated environments	Weapons, Non-weapons
[16]	Scaled-YOLOv4	Custom dataset	Weapons
[17]	YOLOv5 + MobileNetV2	Realistic video	Weapons, Faces, Violence
[18]	YOLO + Pose Estimation	Custom dataset	Weapons + Posture
Our work	YOLOv11	CCTV surveillance videos	People, Weapons, Masks

of intelligent video surveillance.

III. PROPOSED METHOD

In this study, we propose an innovative approach for the automated analysis of images extracted from online video streams, aiming to simultaneously detect the presence of people, facial masks, and weapons. The method relies on an advanced object detection architecture, **YOLOv11**, optimized for real-time processing and multi-class classification in complex scenarios.

A. Dataset Construction

The initial phase involved the creation of a heterogeneous and representative dataset by integrating both manually generated and publicly available sources.

Manual Extraction: A significant portion of the dataset was built manually by extracting frames from publicly accessible online videos, mainly from surveillance cameras. This subset, referred to as the **Robbery Action Dataset** [22], provided high environmental variability in terms of lighting conditions, image quality, and scenes. Images were manually annotated using the YOLOv11 format via the Roboflow platform, which facilitated accurate and consistent labeling. Given the relatively small size of our custom dataset, we chose not only to apply data augmentation techniques but also to expand the dataset by incorporating dataset open source. These datasets were carefully selected and re-annotated to ensure consistency with our target class definitions and annotation format. This strategy allowed us to significantly increase data diversity while maintaining a coherent and unified dataset structure.

Open Source Datasets The remaining part of the dataset was derived from open-source databases available on platforms such as Roboflow and GitHub. These datasets were standardized and filtered to include only the three target classes: *person*, *mask*, and *weapon*. In fact, all annotations were converted to the YOLOv11 format, using the same class index convention (0 = mask, 1 = person, 2 = weapon). Only the three target classes -*person*, *mask*, and

Table II. NUMBER OF ANNOTATED INSTANCES PER CLASS ACROSS THE TRAIN, VALIDATION, AND TEST SETS.

Class	Train (78.5%)	Validation (11.5%)	Test (11.5%)	Total
Mask	11425	1223	723	13371
Person	5839	855	777	7471
Weapon	3214	493	350	4057
Total	20478	2571	1850	24899

weapon - were retained. All other labels were removed. Inconsistent class naming (e.g., *gun*, *knife*,) was unified under a single class: *weapon*.

The datasets used include:

- **Face Mask Detection dataset** [23]
- **Weapon Recognition dataset** [24]
- **Person Detection dataset** [25]

All images in these datasets were already labeled. In some cases, manual review or re-annotation was performed to ensure consistency between formats and class definitions.

Final Dataset Composition: The resulting dataset consists of **7,986 JPG images**, divided as follows:

- **Training set:** 6267 images (78.5%)
- **Validation set:** 880 images (11%)
- **Test set:** 839 images (10.5%)

In addition to reporting the number of images per dataset, Table II shows the total number of annotated object instances for each class (*person*, *mask*, *weapon*) across the training, validation, and test splits.

This breakdown offers a clearer view of the actual object distribution, which reveals a certain class imbalance - particularly a higher number of *mask* annotations. This was addressed through targeted data augmentation and oversampling strategies during training.

B. Preprocessing and Data Augmentation

To ensure input uniformity and enhance the model's generalization capabilities, the following preprocessing steps were applied:

- Automatic orientation correction (removal of EXIF metadata)
- Resizing images to 640×640 pixels with proportional stretching

Subsequently, targeted data augmentation techniques were implemented to increase image diversity:

- Random rotation between -9° and $+10^\circ$
- Random brightness variation ($\pm 15\%$)
- Random Gaussian blur between 0 and 1 pixel
- Horizontal and vertical shifting between -10% and $+10\%$
- Application of salt and pepper noise to 0.2% of pixels

These augmentation techniques were also used to address class imbalance by oversampling underrepresented classes. This not only improved class distribution within

the training set but also contributed to mitigating overfitting and enhancing the model's performance on the test set.

C. YOLOv11 Architecture

In our work, we use the YOLOv11 algorithm for its high performance in terms of accuracy and speed [26]. The adopted configuration allows for the simultaneous detection of target classes, ensuring effective handling of partial occlusions.

D. Objectives and Method Flexibility

The primary objective of the proposed method is to provide an automated solution for the timely detection of people, face masks, and weapons within surveillance contexts, such as public safety, urban monitoring, and the surveillance of sensitive environments. Nonetheless, this study also emphasizes the importance of modularity and flexibility, enabling effective adaptation to future scenarios characterized by emerging requirements or changes in operational contexts. Key features that make the system easily extensible include:

- *Class integration:* the dataset structure and network architecture allow for the addition of new classes;
- *Compatibility with new video sources:* the system can be easily adapted to videos from different sources, subject to input format normalization.

IV. EXPERIMENTAL RESULTS

The training process was conducted using the YOLOv11n model, consisting of 181 layers and approximately 2.59 million parameters, with a computational complexity of 6.4 GFLOPs. The model was trained for 100 epochs on a dataset comprising 6,259 valid training images and 873 validation images, all resized to 640×640 pixels. Automatic optimizer selection by the Ultralytics engine resulted in the adoption of the AdamW optimizer, with an effective learning rate of 0.001429 and a momentum of 0.9. Data augmentation strategies included horizontal flipping, random translation, and random erasing, along with RandAugment policies. Training was performed using mixed precision (AMP) on an NVIDIA A100 GPU with 40 GB of VRAM, using PyTorch 1.13.1 and Ultralytics YOLO version 8.3.169. The training pipeline also included a warm-up phase over the first 3 epochs and utilized a weight decay of 0.0005 to improve generalization. In the final testing phase, a confusion matrix was computed (Figure 1) to evaluate the model's performance on a dataset never seen during the training phase, thus assessing its ability to generalize; each row represents the normalized distribution of predicted classes for a given real class label.

The results show that the model achieved an accuracy of 65.35% in recognizing the target classes on the test set. The weighted precision, equal to 71.94%, suggests a good ability of the model to avoid false positives,

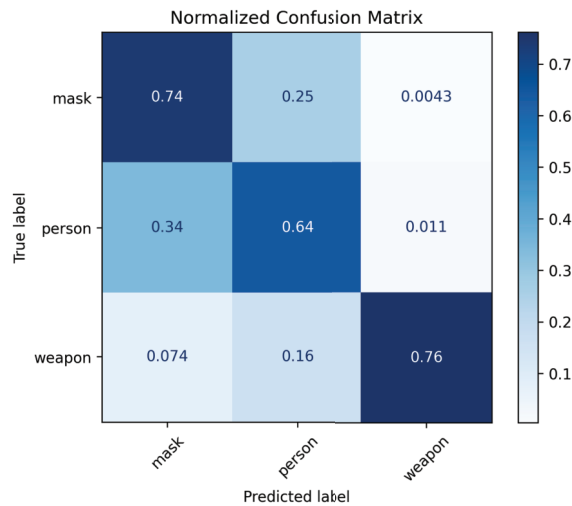


Figure 1. Normalized confusion matrix. Each row represents a true class (ground truth), and each column represents the class predicted by the model. Values indicate the proportion of instances of each true class assigned to each predicted class.

especially for imbalanced classes. However, the weighted recall (65.35%) highlights that there is still room for improvement in identifying all relevant instances. The weighted F1 score, which represents a balance between precision and recall, reached 67.97%, confirming a moderately solid yet improvable performance. These values indicate that the approach is promising, although further optimization would be beneficial, particularly to enhance sensitivity toward underrepresented classes.

Table III. EVALUATION METRICS ON THE TEST SET.

Metric	Value
Accuracy	65.35%
Weighted Precision	71.94%
Weighted Recall	65.35%
Weighted F1 Score	67.97%

The Figure 2 shows a real scene extracted from the test set, where a person is caught committing a robbery inside a store. The model, based on the YOLOv11 architecture, successfully identifies the person and the weapon, as well as the absence of a face mask, confirming the effectiveness of our approach in detecting and classifying objects in complex and dynamic scenarios. The correct labeling of the interaction between the person and the weapon represents important empirical evidence of the system’s capability to support surveillance and crime prevention activities in critical environments.

V. DISCUSSION AND LIMITATIONS

The proposed system demonstrates how the integration of recent advances in object detection - specifically the YOLOv11 architecture - can be effectively adapted and deployed in realistic, high-complexity video surveillance scenarios. Unlike previous studies that often rely on

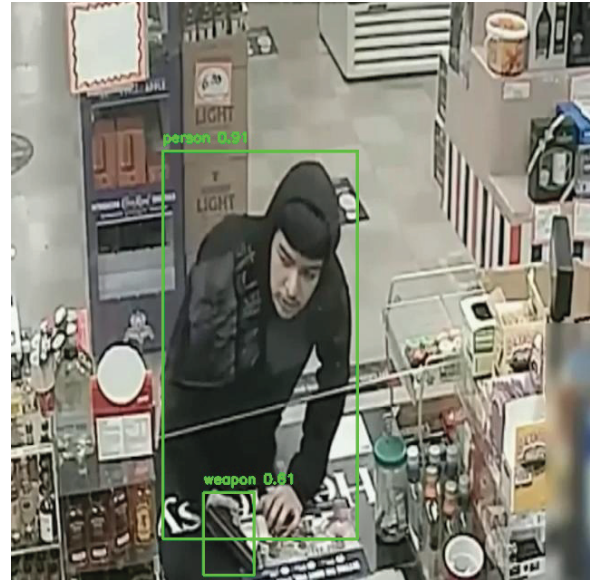


Figure 2. Detection example on the test set.

curated or synthetic datasets, we constructed and used a dataset directly derived from real-world CCTV footage, ensuring a much higher degree of realism in terms of lighting variability, motion artifacts, occlusions, and background clutter.

Furthermore, this work introduces a multi-class detection pipeline capable of simultaneously identifying people, weapons, and identity-concealing items (such as masks). This combination enables richer scene understanding and paves the way for the development of smarter, context-aware surveillance systems. Prior literature typically addresses these classes in isolation or within simplified scenarios. By contrast, we focus on joint, real-time detection in uncontrolled environments—an under-explored yet practically critical area.

Thus, this study contributes to applied research by (i) validating the use of YOLOv11 in complex, real-time video contexts; (ii) demonstrating its effectiveness for joint detection of multiple threat-related classes; and (iii) highlighting the importance of realistic datasets for performance benchmarking.

A key challenge in this work was the design and construction of a dedicated dataset tailored to the application scenario. This represents a significant engineering contribution: the dataset was built entirely from scratch using real surveillance footage, with the aim of capturing complex, multi-object, and context-rich scenes. One limitation currently observed involves the distinction between the “person” and “mask” classes, as highlighted by the confusion matrix (Figure 1). These misclassifications primarily reflect the current limitations of the dataset. Despite efforts to mitigate this through augmentation, the imbalance remains a critical issue. We are addressing this by

extending the dataset to include a wider range of samples, thereby enabling better generalization and representation of edge cases. We are confident that the combination of a richer dataset and more sophisticated models will result in a significant reduction in misclassification rates and improved system performance for real-time security tasks. While the present study focused on validating feasibility and dataset construction, we recognize that comparative evaluation against alternative architectures - including other YOLO versions (e.g., YOLOv5, YOLOv8), and Faster R-CNN) - is essential to objectively assess the strengths and weaknesses of our approach. To further improve performance and address these limitations, we are working on extending the dataset with new data from actual surveillance scenarios. The expanded dataset will incorporate greater diversity in scenes, lighting conditions, and challenging cases such as occlusions and crowded environments. This will enhance the model's generalization ability and improve the robustness of multi-class detection in unconstrained real-world contexts.

VI. CONCLUSION AND FUTURE WORK

In this study, we proposed a novel approach for the simultaneous detection of persons, face masks, and weapons in images extracted from real-world surveillance videos, leveraging the YOLOv11 architecture. The system demonstrated promising results (precision: 71.94%, recall: 65.35%, F1-score: 67.97%, accuracy: 65.35%), despite the challenges posed by partial occlusions and the complexity of multi-object scenes. A key contribution of this work is the construction of a custom dataset derived from actual CCTV footage, which better reflects the conditions of operational video surveillance compared to conventional datasets. This dataset captures complex and realistic scenes, enabling multi-class object detection (people, weapons, masks) under varying lighting conditions and image quality levels. The dataset developed in this study represents a distinctive element compared to other studies, which often rely on isolated and decontextualized images focused solely on the presence of weapons or individuals. Our method performs object detection simultaneously for weapons, persons, and masks within images derived from surveillance cameras. This methodological choice enabled the development of a model more closely aligned with operational environments, although annotation quality and data consistency still represent areas for improvement. In this regard, a complete revision of the dataset is proposed, with manual annotation through specialized platforms such as *Roboflow*, in a YOLOv11-compatible format.

Future work include: the generation of synthetic images through Generative Adversarial Networks (GANs), to enrich the dataset with realistic yet artificial scenes simulating complex events such as robberies, threats, and altercations, thereby improving the model's generalization capability in rare but critical scenarios; the refinement of annotations for better class representation; and the

extension of classification tasks to facial analysis, to differentiate between masked and unmasked individuals in potentially critical situations.

Furthermore, we plan to conduct a systematic evaluation of these models on our dataset to determine the most suitable architecture to ambiguous or occluded conditions. This comparative analysis will help refine model selection and develop a more efficient and reliable surveillance system.

In conclusion, this work lays the foundation for an intelligent and proactive monitoring system that can be further enhanced through dataset improvement strategies and deep learning model optimization, with the ultimate goal of strengthening public safety in real-world environments.

ACKNOWLEDGMENT

This research was partially supported by the VIA SICURA (Visione e Intelligenza Artificiale per la Sicurezza delle persone in ambito URbano e lavorativo e dell'Ambiente) project, funded by MIMIT, no. F/360174/01/X75.

DECLARATION ON GENERATIVE AI

During the preparation of this work, the authors used ChatGPT to paraphrase and refine the text, enhancing its fluency and coherence. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

REFERENCES

- [1] L. He, S. Wen, L. Wang, and F. Li, "Vehicle theft recognition from surveillance video based on spatiotemporal attention," *Applied Intelligence*, vol. 51, pp. 2128–2143, 2021.
- [2] G. Morales, I. Salazar-Reque, J. Telles, and D. Díaz, "Detecting violent robberies in cctv videos using deep learning," in *IFIP International Conference on Artificial Intelligence Applications and Innovations*. Springer, 2019, pp. 282–291.
- [3] R. Kakadiya, R. Lemos, S. Mangalan, M. Pillai, and S. Nikam, "Ai based automatic robbery/theft detection using smart surveillance in banks," in *2019 3rd International conference on Electronics, Communication and Aerospace Technology (ICECA)*. IEEE, 2019, pp. 201–204.
- [4] W. Sultani, C. Chen, and M. Shah, "Real-world anomaly detection in surveillance videos," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 6479–6488.
- [5] R. Vrskova, R. Hudec, P. Kamencay, and P. Sykora, "A new approach for abnormal human activities recognition based on convlstm architecture," *Sensors*, vol. 22, no. 8, p. 2946, 2022.
- [6] A. Jain and D. K. Vishwakarma, "State-of-the-arts violence detection using convnets," in *2020 international conference on communication and signal processing (ICCSPP)*. IEEE, 2020, pp. 0813–0817.
- [7] K. Kheta, C. Delgove, R. Liu, A. Aderogba, M.-O. Pokam, M. M. Unal, Y. Xing, and W. Guo, "Vision-based conflict detection within crowds based on high-resolution human pose estimation for smart and safe airport," *arXiv preprint arXiv:2207.00477*, 2022.
- [8] M. Baba, V. Gui, C. Cernazanu, and D. Pescaru, "A sensor network approach for violence detection in smart cities using deep learning," *Sensors*, vol. 19, no. 7, p. 1676, 2019.
- [9] D. Won, Z. C. Steinert-Threlkeld, and J. Joo, "Protest activity detection and perceived violence estimation from social media images," in *Proceedings of the 25th ACM international conference on Multimedia*, 2017, pp. 786–794.

- [10] S. Motiian, F. Siyahjani, R. Almohsen, and G. Doretto, "Online human interaction detection and recognition with multiple cameras," *IEEE transactions on Circuits and Systems for Video Technology*, vol. 27, no. 3, pp. 649–663, 2016.
- [11] M. Komar, V. Kochan, L. Dubchak, A. Sachenko, V. Golovko, S. Bezobrazov, and I. Romanets, "High performance adaptive system for cyber attacks detection," in *2017 9th IEEE international conference on intelligent data acquisition and advanced computing systems: technology and applications (IDAACS)*, vol. 2. IEEE, 2017, pp. 853–858.
- [12] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 779–788.
- [13] P. Yadav, N. Gupta, and P. K. Sharma, "A comprehensive study towards high-level approaches for weapon detection using classical machine learning and deep learning methods," *Expert Systems with Applications*, vol. 212, p. 118698, 2023.
- [14] K. Vijayakumar, K. Pradeep, A. Balasundaram, and A. Dhande, "R-cnn and yolov4 based deep learning model for intelligent detection of weaponries in real time video," *Mathematical Biosciences and Engineering*, vol. 20, no. 12, pp. 21 611–21 625, 2023.
- [15] A. Thakur, A. Shrivastav, R. Sharma, T. Kumar, and K. Puri, "Real-time weapon detection using yolov8 for enhanced safety," *arXiv preprint arXiv:2410.19862*, 2024.
- [16] S. Ahmed, M. T. Bhatti, M. G. Khan, B. L'ovstr'om, and M. Shahid, "Development and optimization of deep learning models for weapon detection in surveillance videos," *Applied Sciences*, vol. 12, no. 12, p. 5772, 2022.
- [17] M. M. Mukto, M. Hasan, M. M. Al Mahmud, I. Haque, M. A. Ahmed, T. Jabid, M. S. Ali, M. R. A. Rashid, M. M. Islam, and M. Islam, "Design of a real-time crime monitoring system using deep learning techniques," *Intelligent Systems with Applications*, vol. 21, p. 200311, 2024.
- [18] A. R. Maligireddy, M. R. Uppula, N. Rastogi, and Y. R. Parla, "Gun detection using combined human pose and weapon appearance," *arXiv preprint arXiv:2503.12215*, 2025.
- [19] A. Velasco-Mata, "Human pose information as an improvement factor for handgun detection," Master's thesis, Escuela Superior de Informatica (UCLM), 2020.
- [20] J. Ruiz-Santaquiteria, A. Velasco-Mata, N. Vallez, G. Bueno, J. A. Alvarez-Garcia, and O. Deniz, "Handgun detection using combined human pose and weapon appearance," *IEEE Access*, vol. 9, pp. 123 815–123 826, 2021.
- [21] J. Ruiz-Santaquiteria, A. Velasco-Mata, N. Vallez, O. Deniz, and G. Bueno, "Improving handgun detection through a combination of visual features and body pose-based data," *Pattern Recognition*, vol. 136, p. 109252, 2023.
- [22] R. Avanzato, "Robbery action dataset," <https://universe.roboflow.com/roberta-avanzato/robbery-action>, apr 2025, visited on 2025-07-25. [Online]. Available: <https://universe.roboflow.com/roberta-avanzato/robbery-action>
- [23] D. Ninja, "Visualization tools for face mask detection dataset," <https://datasetninja.com/face-mask-detection>, jul 2025, visited on 2025-07-25. [Online]. Available: <https://datasetninja.com/face-mask-detection>
- [24] Morgan12003, "Weapon recognition model," <https://github.com/Morgan12003/Weapon-recognition-model>, 2023, accessed: 2025-07-25.
- [25] Learning, "Person detection dataset," <https://universe.roboflow.com/learning-fkrr5/person-detection-mjjvf>, dec 2023, visited on 2025-07-25. [Online]. Available: <https://universe.roboflow.com/learning-fkrr5/person-detection-mjjvf>
- [26] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements. arxiv 2024," *arXiv preprint arXiv:2410.17725*.